LESSON PLAN

COMPUTER (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to communication system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to communication system. Purpose, technical details and types of various communication systems.
5. Specific Objectives	26.1.1.1 Trainee will be able to understand the basic theory of communication.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of basic signal transmission diagrams; Safety: No live equipment handling; Instructor supervision: Monitor understanding of theory only.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of a communication system?
(b) Define the term 'signal' in the context of communication.
(c) List two types of communication systems based on transmission medium.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic theory of communication • Purpose of communication systems • Key components of a system • Signal transmission process	L & Ds	PPT, IB	15 min
(b)	• Technical details of analog systems • Technical details of digital systems • Wired vs wireless transmission • Modulation techniques overview	L & Ds	PPT, PDF	20 min
(c)	• Classification of communication systems • Terrestrial vs satellite communication • Fiber optic basics • Radio frequency spectrum usage	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between analog and digital communication systems.
(b) Describe the role of modulation in a communication system.
(c) Identify three types of communication systems based on the transmission medium.

12. Home Assignment and Follow Up	Prepare a brief note on the safety precautions required when handling embedded system components, to be discussed in LP 08.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Safety precautions while working on Embedded System	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min